The problem

Relief happens where the signal isn’t.

9,226 people

were paid house grants twice after Cyclone Amphan: **₹18 crore**. Every camp’s list was fine on its own. The mistake only appeared once the lists were put together.

About 10,000

Indian villages have **no mobile signal at all**.

Google Sheets

held the missing-persons list after the Wayanad landslide, updated every day. It **needs the internet**.

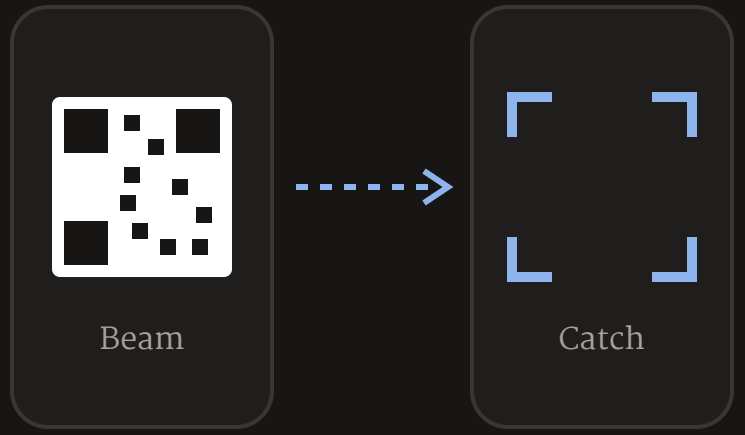
Amphan: The Federal, on the government auditors’ report · Villages: Free Press Journal · Wayanad: South First

What we built

Relay Docs

Your doc is passed hand to hand, like a relay.

- A document editor that runs in any phone's browser. Nothing to install.
- Every phone carries the whole doc.
- When two phones meet, one shows a moving code and the other's camera catches it. No internet, no Wi-Fi, no Bluetooth. A small file works too.
- They can meet in any order, and everyone ends up with the same doc.



The truck back: a picture and three words made from what the doc says, so anyone can see two phones hold the same doc.

What’s new

Three things we couldn’t find in any document editor

1

Both versions kept

When two people rewrite the same sentence apart, both versions stay, the same on every phone.

Rina, Gosaba camp
Gosaba ferry runs at 6 am.

Asif, Patharpratima camp
No ferry to Gosaba today.

2

Rules inside the doc

Checked the same way on every phone, whatever order they synced in. That’s the Amphan mistake, caught.

One kit per ration card. Ration card 4471 has Family kit 2 times.

3

Which phones have it

Under every paragraph, which phones already have it. Like blue ticks, with no network.

✓✓ Rina, Asif and Base camp have this

Proof

The judge picks the order.

A normal sync keeps the version that arrived last

> Rina, Gosaba camp, Monday 6:05 am
> Gosaba ferry runs at 6 am. Bring ration cards to the jetty.

Relay Docs keeps both

> Rina, Gosaba camp, Monday 6:05 am
> Gosaba ferry runs at 6 am. Bring ration cards to the jetty.

> Asif, Patharpratima camp, Monday 4:10 pm
> No ferry to Gosaba today. The jetty is under water; wait for the relief boat at 2 pm.

- A normal sync would have sent families to a flooded jetty.
- One tap settles every clash, and nothing is thrown away.
- Every screen shows the same truck back.
- A thousand shuffled orders all come out the same. Run it yourself: `npm run selftest`

Honest about what exists. Every piece exists somewhere: animated codes in Bitcoin wallets, carry-it-forward sync in Briar, replay in Etherpad, kept conflicts in Pijul, fingerprints in Signal, rule-keeping in database research. We couldn’t find anyone who has put them together into a phone document editor that works with no signal at all.

What’s next

We’re starting with relief camps.

Ditto raised 82 million dollars in 2025 to sell offline sync to airlines.

- A signing key on every phone, so no phone can write under someone else’s name.
- Face to face, send only what’s new instead of the whole doc.
- A native Android app that syncs by itself when phones are near each other.

Any doc, any order, no signal.

ranvir7123.github.io/phir-milenge

Open it once with internet. After that it works offline.

Ditto: SiliconANGLE, 12 March 2025